# MAKI Cheat Sheet

Media Asset Keeper & Indexer — v4.5.7

## Global Options

| | |
|---|---|
| `--json` | JSON output to stdout |
| `-v, --verbose` | Show operational details |
| `-l, --log` | Per-file progress to stderr |
| `-d, --debug` | External tool debug output |
| `-t, --time` | Show elapsed time |

## Setup

**`maki init`**
Create a new catalog in the current directory.

**`maki volume add [LABEL] PATH`**
Register a storage volume (label auto-derived from path if omitted).
`--purpose media|working|archive|backup|cloud`

**`maki volume list`**
Show registered volumes with online/offline status.
`--purpose PURPOSE --offline --online`

**`maki volume set-purpose / rename / remove`**
Change purpose, rename, or unregister a volume.

**`maki volume combine / split`**
Merge two volumes or split a subdirectory into new volume.

## Ingest & Edit

**`maki import PATHS...`**
Import files into the catalog.
`--volume --profile NAME --auto-group --smart`
`--embed` [Pro] `--describe` [Pro] `--dry-run`
`--include GROUP --skip GROUP`

**`maki delete ASSET_IDS...`**
Delete assets. `--apply --remove-files`

**`maki tag ASSET_ID TAGS...`**
Add or remove tags. `--remove`

**`maki tag rename OLD NEW`**
Rename a tag across all assets. `--apply`
OLD accepts **=** (exact level) and **^** (case-sensitive) prefixes.

**`maki tag delete TAG`**
Delete a tag (cascades to descendants) from every asset. `--apply`
TAG accepts the same **=** / **^** markers as **`tag rename`**.

**`maki tag clear ASSET_ID`**
Remove all tags from an asset.

**`maki tag fix-unicode`**
NFC-normalise every tag value catalog-wide; merges NFC/NFD duplicates per asset. `--apply`

**`maki tag export-vocabulary`**
Export tag tree as vocabulary.yaml.
`--prune --default --output FILE --counts`
`--format yaml|text|json` (text = tab-indented for Lightroom / Capture One; json = nested objects for scripts; `--counts` adds '# N assets' trailing YAML comments)

**`maki edit ASSET_ID`**
Edit metadata fields.
`--name --description --rating N`
`--label COLOR --date --clear-FIELD`

**`maki group HASHES...`**
Group variants into one asset.

**`maki split ASSET_ID --variant HASH`**
Split variants out of an asset.

**`maki auto-group [QUERY]`**
Auto-group by filename stem (directory-local). `--apply --global`

**`maki describe [QUERY]`** [Pro]
Generate VLM image descriptions.
`--asset ID --volume --model`
`--mode describe|tags|both`
`--apply --force --dry-run`

**`maki auto-tag [QUERY]`** [Pro]
AI auto-tagging with SigLIP.
`--asset --volume --threshold`
`--labels FILE --apply --download`

**`maki embed [QUERY]`** [Pro]
Generate image embeddings.
`--asset --volume --force --export`

## Organize

**`maki collection create / add / remove / list / show / delete`**
Static albums. **`create NAME`**, **`add NAME IDS...`**, **`remove NAME IDS...`**

**`maki saved-search save / list / run / delete`**
Dynamic smart albums. **`save NAME QUERY`** `--sort --favorite`

**`maki stack create ASSET_IDS...`**
Create a stack. First = pick.

**`maki stack add REF ASSET_IDS...`**
Add assets to existing stack.

**`maki stack remove / pick / dissolve / list / show`**
Stack management commands.

**`maki stack from-tag PATTERN`**
Convert tag patterns to stacks. `--apply --remove-tags`

**`maki faces detect / cluster / clean / name / merge`** [Pro]
Face detection and recognition.
`--query --asset --apply --force --threshold --min-confidence`
Diagnostics: **`faces similarity`**, **`faces dump-aligned`**, **`faces status`**

## Retrieve

**`maki search QUERY`**
Search assets with filters (see back).
`--format ids|short|full|json|{template}`
`-q` (shorthand for **`--format ids`**)

**`maki show ASSET_ID`**
Show full asset details.

**`maki preview ASSET_ID`**
Open preview in OS default image viewer.

**`maki export QUERY TARGET`**
Export to directory. `--zip --layout flat|mirror`
`--all-variants --sidecars`

**`maki contact-sheet QUERY OUTPUT`**
Generate PDF contact sheet.
`--layout --paper --landscape`
`--title --fields --sort`

**`maki duplicates`**
Find duplicate files. `--mode same|cross --volume --format`

**`maki stats`**
Catalog statistics breakdown. `--tags --verified`

**`maki status`**
Health overview with next-step hints. `--min-copies N`

**`maki backup-status`**
Show backup coverage and at-risk assets. `--copies N`

**`maki doc [manual|cheatsheet|filters]`**
Open documentation PDF in browser.

**`maki licenses`**
Show MAKI license + 3rd-party crate licenses. `--summary`

**`maki serve`**
Start web UI. `--port --bind`

**`maki shell`**
Interactive shell. `-c CMD --script FILE`

## Maintain

**`maki verify [PATHS...]`**
Check file integrity.
`--volume --asset --force --max-age DAYS`

**`maki sync [PATHS...]`**
Sync with disk changes. `--apply --remove-stale`

**`maki sync-metadata [QUERY]`** [Pro]
Bidirectional XMP sync. `--asset --dry-run`

**`maki refresh [PATHS...]`**
Re-read changed recipes. `--dry-run --media`
`--reimport` — full re-extract `--exif-only` — EXIF only

**`maki writeback [QUERY]`** [Pro]
Write pending edits to XMP. `--all --dry-run`

**`maki cleanup`**
Remove stale records. `--apply --list --path`

**`maki dedup`**
Remove same-volume duplicate file locations. `--apply --prefer`

**`maki generate-previews [PATHS...]`**
Generate/regenerate previews.
`--force --smart --upgrade`

`--volume --asset --include --skip`

**`maki relocate`** *`[ASSET_IDS...]`*
Copy/move files between volumes.

`--target VOLUME --query --remove-source`
`--create-sidecars --dry-run`

**`maki rebuild-catalog`**
Rebuild SQLite from YAML sidecars. `--asset ID` (per-asset)

Also: `migrate`, `create-sidecars`, `update-location`, `fix-roles`, `fix-dates`, `fix-recipes`

# MAKI Cheat Sheet

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## Core Concepts

| | |
|---|---|
| **Asset** | Logical media item (one photo) |
| **Variant** | Concrete file (Original, Alternate, Processed, Export) |
| **Recipe** | XMP/processing sidecar (tracked by location) |
| **Volume** | Storage device (online/offline detection) |
| **Stack** | Collapsed group in browse (own metadata) |
| **Collection** | Static album (manual curation) |
| **Saved Search** | Dynamic smart album (query re-evaluated) |

## Key Workflows

### Import & Organize

1. `maki init` + `maki volume add`
2. `maki import --smart /path`
3. `maki serve` — browse in web UI
4. Rate, tag, label via UI or CLI
5. `maki auto-tag ""` [Pro]— AI tagging

### Maintenance Cycle

1. `maki verify` — check integrity
2. `maki sync` — detect moved/missing
3. `maki sync-metadata` — bidirectional XMP [Pro]
4. `maki cleanup --apply` — remove stale

### Backup & Archive

1. `maki volume add Archive /mnt/nas --purpose backup`
2. `maki relocate --target Archive ""` — copy all
3. `maki backup-status` — check coverage
4. `maki dedup --apply` — remove same-volume dups

### Export & Deliver

1. `maki export "rating:4+ tag:portfolio" ./out`
2. `maki export "collection:Client" ./out --zip`
3. `maki contact-sheet "rating:5" portfolio.pdf`

## Search Filter Syntax

All filters work in CLI `maki search` and web UI search box.

### Text & Metadata

| | |
|---|---|
| `tag:landscape` | Tag match (any level) |
| `tag:=` / `tag:/` | `=` whole path (exact value); `/` leaf only |
| `tag:^Landscape` | Case-sensitive (not "landscape") |
| `tag:\|wed` | Prefix anchor (wedding, wedding-2024, ...) |
| `-tag:rejected` | Exclude tag |
| `type:image` | Asset type |
| `format:nef` | File format |
| `label:Red` | Color label |
| `camera:fuji` | Camera (partial) |
| `lens:56mm` | Lens (partial) |
| `description:cat` | Description (substring; alias `desc:`) |
| `person:Alice` | Named person (repeat for AND, comma for OR) [Pro] |
| `collection:Fav` | Collection |
| `path:2026/` | Path prefix (`*` = wildcard) |
| `id:72a0` | Asset ID prefix |
| `meta:key=val` | Source metadata |

### Numeric Filters

All support: `x` (exact), `x+` (min), `x-y` (range), `x,y` (OR)

| | |
|---|---|
| `rating:3+` | 3 stars or more |
| `rating:3-5` | 3 to 5 stars |
| `rating:2,4` | Exactly 2 or 4 |
| `rating:0` | Unrated ("the rest") |
| `iso:100-800` | ISO range |
| `focal:35-70` | Focal length range |
| `f:1.4-2.8` | Aperture range |
| `width:4000+` | Min width (px) |
| `height:2000+` | Min height (px) |
| `copies:2+` | File copies |
| `variants:2+` | Variant count |
| `scattered:2+` | Distinct directories (`/N` depth) |
| `duration:30+` | Video duration (seconds) |
| `codec:h264` | Video codec (partial) |
| `faces:2+` | Face count [Pro] |

### Date Filters

| | |
|---|---|
| `date:2026` | Year prefix |
| `date:2026-03` | Month prefix |
| `date:2026-03-15` | Exact date |
| `dateFrom:2026-01` | From (inclusive) |
| `dateUntil:2026-12` | Until (inclusive) |

### Status Filters

| | |
|---|---|
| `orphan:true` | No file locations |
| `orphan:false` | Has file locations |
| `missing:true` | Files missing on disk |
| `stale:30` | Not verified in N days |
| `stacked:true` | In a stack |
| `volume:Archive` | On specific volume |
| `volume:none` | No online volume |
| `geo:any` | Has GPS data |

### Pro Filters

| | |
|---|---|
| `similar:72a0bb4b` | Visual similarity |
| `min_sim:90` | Similarity threshold |
| `text:sunset` | Text-to-image search |
| `embed:none` | No embedding |

### Combining Filters

| | |
|---|---|
| `tag:a,b` | a OR b (comma) |
| `tag:a tag:b` | a AND b (repeat) |
| `-tag:x` | NOT (dash prefix) |

## Output Formats

`maki search --format` *`PRESET|TEMPLATE`*

| | |
|---|---|
| `ids` | One UUID per line |
| `short` | ID, name, type, format, date |
| `full` | + tags, description, rating |
| `json` | JSON array |
| `'{id}\t{name}'` | Custom template |

Placeholders: `{id}`, `{name}`, `{filename}`, `{type}`, `{format}`, `{tags}`, `{description}`, `{rating}`, `{label}`, `{date}`, `{size}`

## Configuration (maki.toml)

| | |
|---|---|
| `[preview]` | max_edge, format, quality, smart_max_edge |
| `[serve]` | port, bind, per_page |
| `[import]` | exclude, auto_tags, smart_previews, profiles |
| `[browse]` | default_filter |
| `[cli]` | log, time, verbose (default global flags) |
| `[writeback]` | enabled (default: false) [Pro] |
| `[group]` | session_root_pattern (auto-group scope) |
| `[dedup]` | prefer |
| `[verify]` | max_age_days |
| `[ai]` | model, threshold, labels [Pro] |
| `[vlm]` | endpoint, model, timeout [Pro] |
| `[contact_sheet]` | layout, paper, fields |

MAKI
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https://maki-dam.com/